

PARTHIBAN GUNASEKARAN

Software Engineer | Java, Python, C#, JavaScript | Distributed Systems at Scale | Live-Site Debugging and Telemetry | Azure
+91 9843777061 | gunasekaranparthiban31@gmail.com | Coimbatore, Tamil Nadu, India | Open to relocate to Bangalore or Hyderabad
linkedin.com/in/parthiban-gunasekaran1159a5282 | leetcode.com/u/Parthiban_G

EDUCATION

B.E. Computer Science (Internet of Things), Sri Krishna College of Technology, Coimbatore | Currently enrolled | CGPA 7.8 / 10 | Expected Oct 2027

Coursework: Data Structures and Algorithms, Operating Systems and Concurrency, Computer Networks, DBMS and query processing, Object Oriented Design

PROFESSIONAL SUMMARY

Software engineer with **production experience on a distributed service running live for customers**. On **NexaRide** I write and review code on a **48-module backend of 368 REST endpoints** over a **77-model PostgreSQL schema**, where a real-time layer **scales horizontally across nodes through a Redis pub/sub bus**. I debug live issues with **logs, crash telemetry and health checks**, contribute to **46 automated test suites and a CI pipeline** gating every change, and keep the service available through **zero downtime deployment** - the operational bar an on-call engineer is held to. Strong fundamentals: **600+ LeetCode problems in Java**, data structures, algorithms, complexity analysis and object oriented design. **Azure AZ-900 certified**. I use AI tools daily and review every AI-generated change before it ships.

TECHNICAL SKILLS

Languages: Java, Python, C#, JavaScript, TypeScript, SQL, Bash

Computer science fundamentals: Data structures, algorithm design, complexity analysis, object oriented design, concurrency and multithreading, operating systems, computer networks

Distributed systems and scale: Horizontally scaled stateless services, Redis pub/sub message bus, WebSocket gateways, event-driven design, caching, idempotent and transactional writes under concurrency, indexing and partitioning for large relational datasets

Debugging, telemetry and quality: Structured logging, crash reporting and error tracking, health checks, latency metrics, production incident debugging, root cause analysis, unit and integration testing, test driven development, code review, static analysis and strict type checking

Cloud, data and infrastructure: Microsoft Azure (AZ-900), AWS (EC2, RDS, S3), Docker, Terraform, GitHub Actions CI/CD, zero downtime deployment, Linux; PostgreSQL with PostGIS, MySQL, Redis, MongoDB, SQL query optimisation

Backend frameworks: Java with Spring Boot, JPA and Hibernate; Node.js with NestJS and Express.js; REST API design, JWT authentication and role-based access control, rate limiting

PROFESSIONAL EXPERIENCE

Software Engineer Intern, Dendo Aug 2026 - Present

NexaRide platform - live on Google Play and the App Store

- **Write maintainable, extensible code:** ship features end to end on a **48-module service-oriented TypeScript backend (34 controllers, 55 services, 368 REST endpoints)** over a **77-model PostgreSQL schema**, following the established layering and design patterns, with **code review on every pull request** in two-week sprints.
- **Distributed systems at scale:** work on a real-time layer that **scales horizontally across server nodes through a Redis pub/sub adapter**, so any instance can serve any client without sticky sessions; kept concurrent assignment correct with transactional and idempotent writes.
- **Debug proactively with logs and telemetry:** trace failures across services using structured logs, boot health checks and crash reporting - added error tracking that surfaced a **class of previously silent client failures** that no existing signal had caught.
- **Testing and automated analysis:** contribute to **46 Jest and Supertest suites** across services and API contracts, and to a path-filtered **GitHub Actions** pipeline that gates every change on lint, static type checks and tests before it can merge.
- **Performance, scalability and system dependencies:** replaced a sequential one-at-a-time assignment algorithm with zone-based distance-band broadcast matching, cutting assignment latency, and tuned slow relational and geospatial queries with targeted indexes.
- **Operational excellence and integration:** services ship as **Docker** images to **Terraform**-provisioned cloud infrastructure with **zero downtime deployment** gated on a boot health check, so a bad build cannot take the service down; integrate across backend, client and infrastructure through a typed shared contract package used by three client applications.

Java Developer Intern, Pinnacle Labs Pvt Ltd Jun 2025 - Sep 2025

- **Java backend development:** built 5 concurrent **Spring Boot** REST services in a controller, service and repository design with **JPA and Hibernate** over MySQL for real-time seat availability and booking, using transactional writes and locking to keep concurrent bookings correct.
- **Performance and complexity analysis:** profiled slow endpoints, traced the cost to 4 inefficient SQL joins and added composite indexes, reducing API response time by about **40 percent** under concurrent load.

PROJECTS

- **Automated Walnut Quality Inspection System** | Python, OpenCV, NumPy, Raspberry Pi | Real-time data processing pipeline that analyses UV image streams to detect contamination, classify quality grades and drive connected hardware. **Paper under peer review.**
- **TrustLance, Decentralized Freelance Marketplace** | Node.js, React.js, Solidity, Web3.js | Backend API, escrow smart contracts and a keyword-matching recommendation algorithm. **Hackathon winning build.**

ACHIEVEMENTS AND CERTIFICATIONS

- Solved **600+ LeetCode problems in Java** across dynamic programming, graphs, trees, binary search, backtracking and hash tables; 50 Days Badge 2026.
- **3x Hackathon Winner:** HackSprint 2026 (iamneo / NIIT Venture), GDG KSR Web3 Hackathon (Shardeum), Pivot Challenge 2025 (Bharati Vidyapeeth).
- **Certifications:** Microsoft Azure **AZ-900**, Oracle Cloud AI Foundations, Cisco CCNA; **published research** in Edge-IoT vision.